

LAB 6

REG NO C027-01-2727/2024

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Exercise 2

Q1. Why is it important to validate email format on the backend?

Answer:

Backend email validation ensures that only correctly formatted email addresses are stored in the database. It prevents invalid data, improves data quality, and provides security because frontend validation can be bypassed.

Q2. What would happen if a product is created with a `supplier_id` that doesn't exist?

Answer:

The database will reject the insertion because the foreign key constraint is violated. The API should return an appropriate error message such as "Supplier not found" or a 404/409 response.

Exercise 3

Q1. How would you handle a scenario where some products would fall below the minimum price?

Answer:

I would skip those products, update only the valid ones, and return a summary showing which products were updated and which were skipped.

Q2. Should this endpoint be available to all users or only administrators?

Answer:

Only administrators or authorized inventory managers should have access because changing prices affects business operations and revenue.

Exercise 4

Q1. Why would it be problematic to allow negative stock adjustments without validation?

Answer:

Negative stock could make inventory inaccurate, allow selling products that do not exist, and lead to poor business decisions.

Q2. How would you handle concurrent stock adjustments?

Answer:

I would use database transactions and row-level locking (or optimistic locking) to ensure that only one update is applied at a time, preventing race conditions.

Exercise 5

Q. Why is it important to have a consistent error response format across all endpoints?

Answer:

A consistent error format makes APIs easier to consume, simplifies frontend development, improves debugging, and allows applications to handle errors in a predictable way.

Product Catalog API 1.0.0 OAS 3.1

/openapi.json

default ^	
GET	/categories List Categories
POST	/categories Create Category
GET	/suppliers List Suppliers
POST	/suppliers Create Supplier
GET	/products List Products
POST	/products Create Product
GET	/products/search Search Products
PATCH	/products/bulk-update Bulk Update
PATCH	/products/adjust-stock Adjust Stock
DELETE	/products/{product_id} Delete Product
GET	/products/{product_id} Get Product
PATCH	/products/{product_id} Update Product
Schemas ^	
Category > Expand all object	
CategoryCreate > Expand all object	

ALL ENDPOINTS

Screenshot of creating a category

POST

/categories Create Category

⌵

Parameters

Cancel

Reset

No parameters

Request body required

application/json

⌵

Edit Value | Schema

```
{
  "name": "Tablets",
  "description": "Portable and durable"
}
```

↻

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/categories' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Tablets",
    "description": "Portable and durable"
  }'
```

📄

Request URL

http://127.0.0.1:8000/categories

Server response

Code

Details

201

Response body

```
{
  "name": "Tablets",
  "description": "Portable and durable",
  "id": 1
}
```

📄 Download

SCREENSHOT OF CREATING SUPPLIERS

POST

/suppliers Create Supplier

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

{
 "name": "kasuku Distributors Ltd",
 "contact_person": "jimmy bli",
 "email": "jimmy@techdistributors.co.ke",
 "phone": "+254716876578",
 "is_active": true
}

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  
  'http://127.0.0.1:8000/suppliers' \  
  -H 'accept: application/json' \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "name": "kasuku Distributors Ltd",  
    "contact_person": "jimmy bli",  
    "email": "jimmy@techdistributors.co.ke",  
    "phone": "+254716876578",  
    "is_active": true  
  }'
```

Request URL

http://127.0.0.1:8000/suppliers

Server response

Code

Details

201

Response body

PATCH BULK SCREENSHOT

PATCH

/products/bulk-update Bulk Update

^

Parameters

Cancel

Name	Description
category * required string (query)	1
discount_percent * required number (query)	100

Execute

Clear

Responses

Curl

```
curl -X 'PATCH' \
'http://127.0.0.1:8000/products/bulk-update?category=1&discount_percent=100' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/products/bulk-update?category=1&discount_percent=100
```

Server response

Code	Details
200	Response body <pre>{ "updated_products": 0 }</pre> Response headers <pre>content-length: 22 content-type: application/json date: Thu, 23 Jul 2026 11:34:41 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

SCREENSHOT OF VALIDATION ERROR

POST

/products

Create Product

⌵

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

⌵

Edit Value

Schemas

```
{
  "name": "dell xps 15",
  "description": "short",
  "brand": "Asus",
  "category": "Phones",
  "price": 50,
  "stock": -5,
  "warranty_months": 48,
  "sku": "ABC123",
  "category_id": 1,
  "supplier_id": 1
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/products' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "dell xps 15",
    "description": "short",
    "brand": "Asus",
    "category": "Phones",
    "price": 50,
    "stock": -5,
    "warranty_months": 48,
    "sku": "ABC123",
    "category_id": 1,
    "supplier_id": 1
  }'
```

Request URL

http://127.0.0.1:8080/products

Server response

Code

Details

422

Error: Unprocessable Content

Response body

```
{
  "success": false,
  "status_code": 422,
  "message": "Validation error",
  "timestamp": "2026-07-23T11:30:29.999885",
  "path": "/products",
  "errors": [
    {
      "field": "body.name",
      "message": "Value error, Name must start with a capital letter",
    }
  ]
}
```

SCREENSHOT OF CREATING A PRODUCT

POST /products Create Product

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "HP EliteBook 840 G8",
  "description": "Business laptop with Intel Core i5 processor, 8GB RAM, 512GB SSD and 14-inch Full HD display.",
  "brand": "HP",
  "category": "Laptops",
  "price": 125000,
  "stock": 20,
  "warranty_months": 24,
  "sku": "LAP-HP-0002",
  "category_id": 1,
  "supplier_id": 1
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  http://127.0.0.1:8080/products' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "HP EliteBook 840 G8",
    "description": "Business laptop with Intel Core i5 processor, 8GB RAM, 512GB SSD and 14-inch Full HD display.",
    "brand": "HP",
    "category": "Laptops",
    "price": 125000,
    "stock": 20,
    "warranty_months": 24,
    "sku": "LAP-HP-0002",
    "category_id": 1,
    "supplier_id": 1
  }'
```

Request URL

http://127.0.0.1:8080/products

Server response

Code

Details

201

Response body